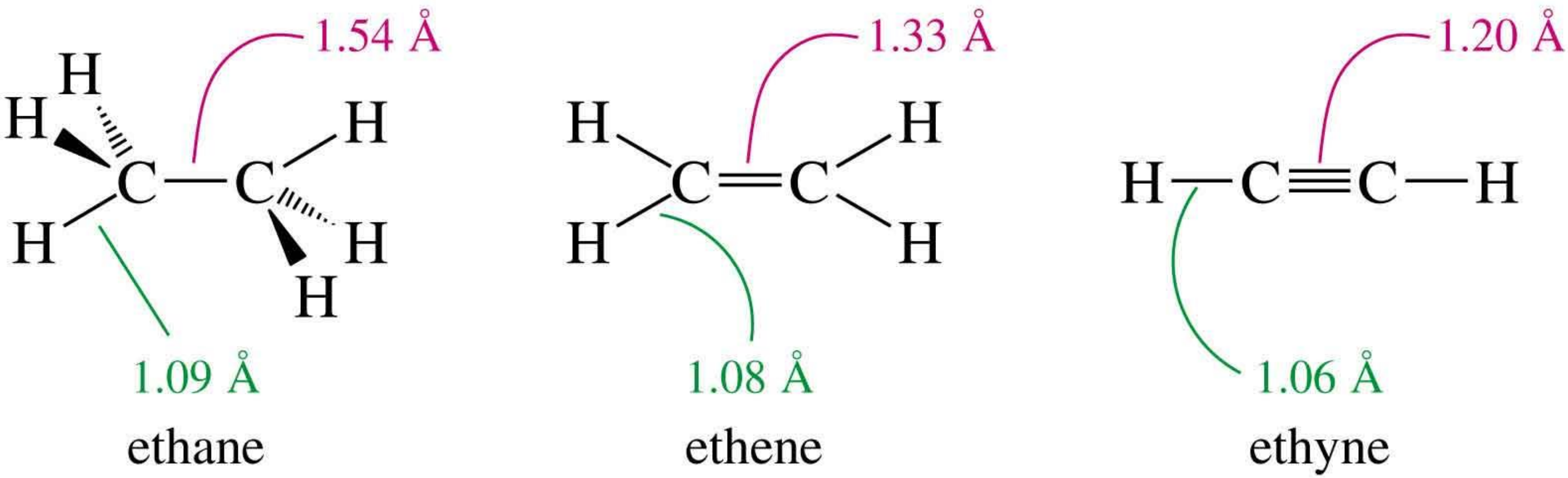
Alkynes

Alkynes

- Alkynes contain one or more triple bond.
- General formula is C_nH_{2n-2}
- Two elements of unsaturation for each triple bond.
- Some reactions are like alkenes: addition and oxidation.
- Nonpolar, insoluble in water.
- Soluble in most organic solvents.
- Boiling points similar to alkane of same size.
- Less dense than water.
- Up to 4 carbons, gas at room temperature.

Bond Lengths

- More *s* (than *p*) character, so shorter length.
- Three bonding overlaps, so shorter.



Bond angle is 180°, so linear geometry. =>

Preparation of Alkynes: Acetylene

- $CaC_2 + KCN + HOH \rightarrow H\text{-}C{\equiv}C\text{-}H + K_2[Cu(CN)_4] + 2KOH$

- Acetylene can be prepared from calcium carbide and water.

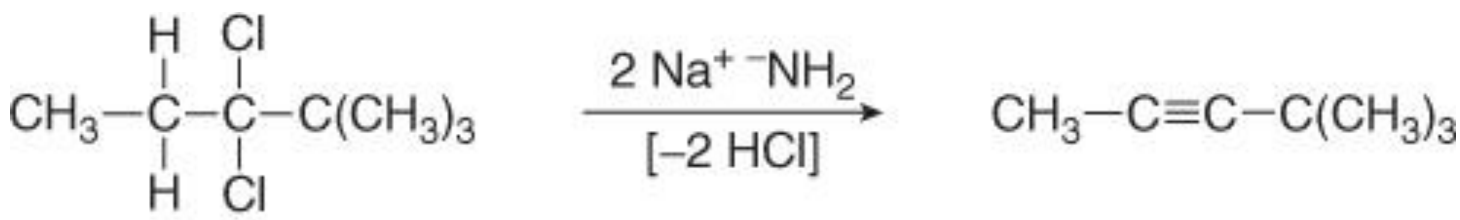
$$CaC_2 + HOH \rightarrow H\text{-}C{\equiv}C\text{-}H$$

- Calcium carbide is prepared by heating coke (Raw Carbon) and calcium oxide in an electric furnace (~2500^o).

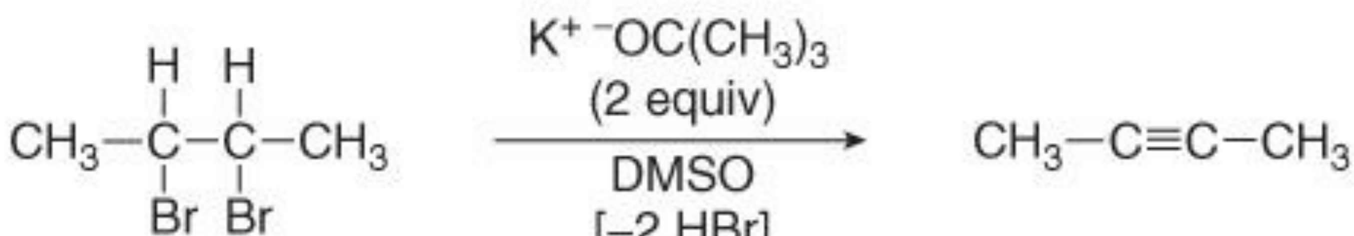
$$C + CaO \rightarrow CaC_2$$

Preparation of Alkynes by Dehydrohalogenation

- alkynes are prepared by elimination reactions. A strong base removes two equivalents of HX from a vicinal or geminal dihalide to yield an alkyne through two successive E2 elimination reactions.



geminal dichloride



vicinal dibromide

- Dehydrohalogenation proceeds in two stages. The removal of 2nd HX molecule is difficulty than removal of 1st HX, requires very strong base and high temperatures.

$$\begin{array}{l} --CX_2CH_2-- \\ --CHXCHX-- \end{array} \xrightarrow{\text{Fast}} --CX=CX-- \xrightarrow{\text{Slow}} --C\equiv C--$$

- Internal alkynes may be converted into terminal alkynes by $NaNH_2$ at 150°C (H+ can be removed from a terminal alkyne by sodium amide, $NaNH_2$).

$$CH_3-C\equiv C-CH_3 + NaNH_2 \xrightarrow{150^0C} CH_3-CH_2-C\equiv C{:}^-\,Na^+ + NH_3$$

$$\downarrow H^+$$

$$CH_3-CH_2-C\equiv CH$$

Reactions of Alkynes

1. Addition reactions
2. Substitution reactions
3. Oxidation reactions

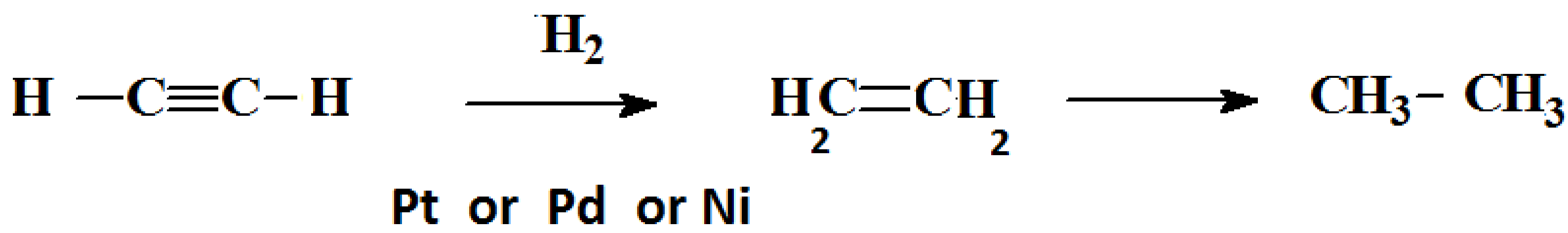
1. Addition reactions

As a general rule, alkynes react with electrophilic reagents in Manner similar to that of alkenes

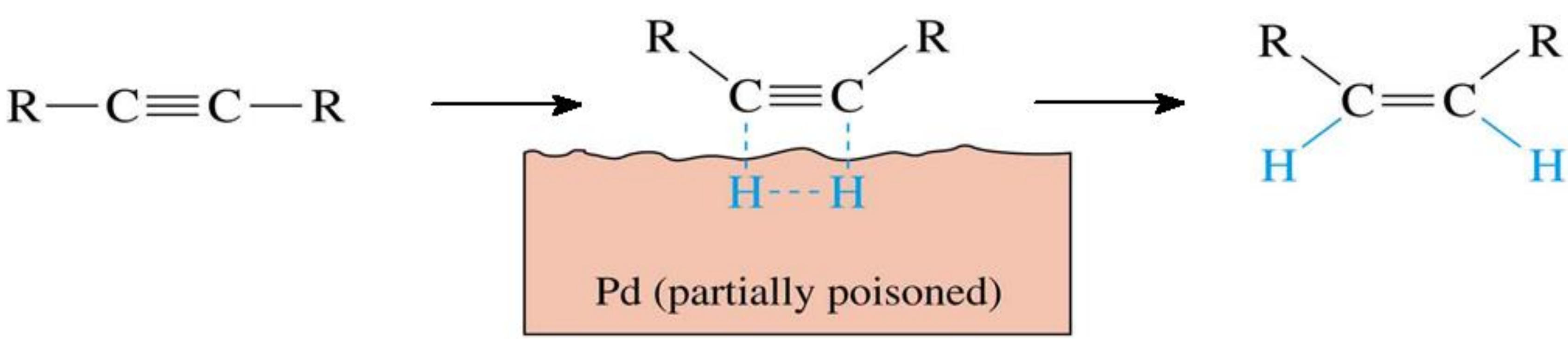
Addition of Hydrogen (Hydrogenation)

Three reactions:

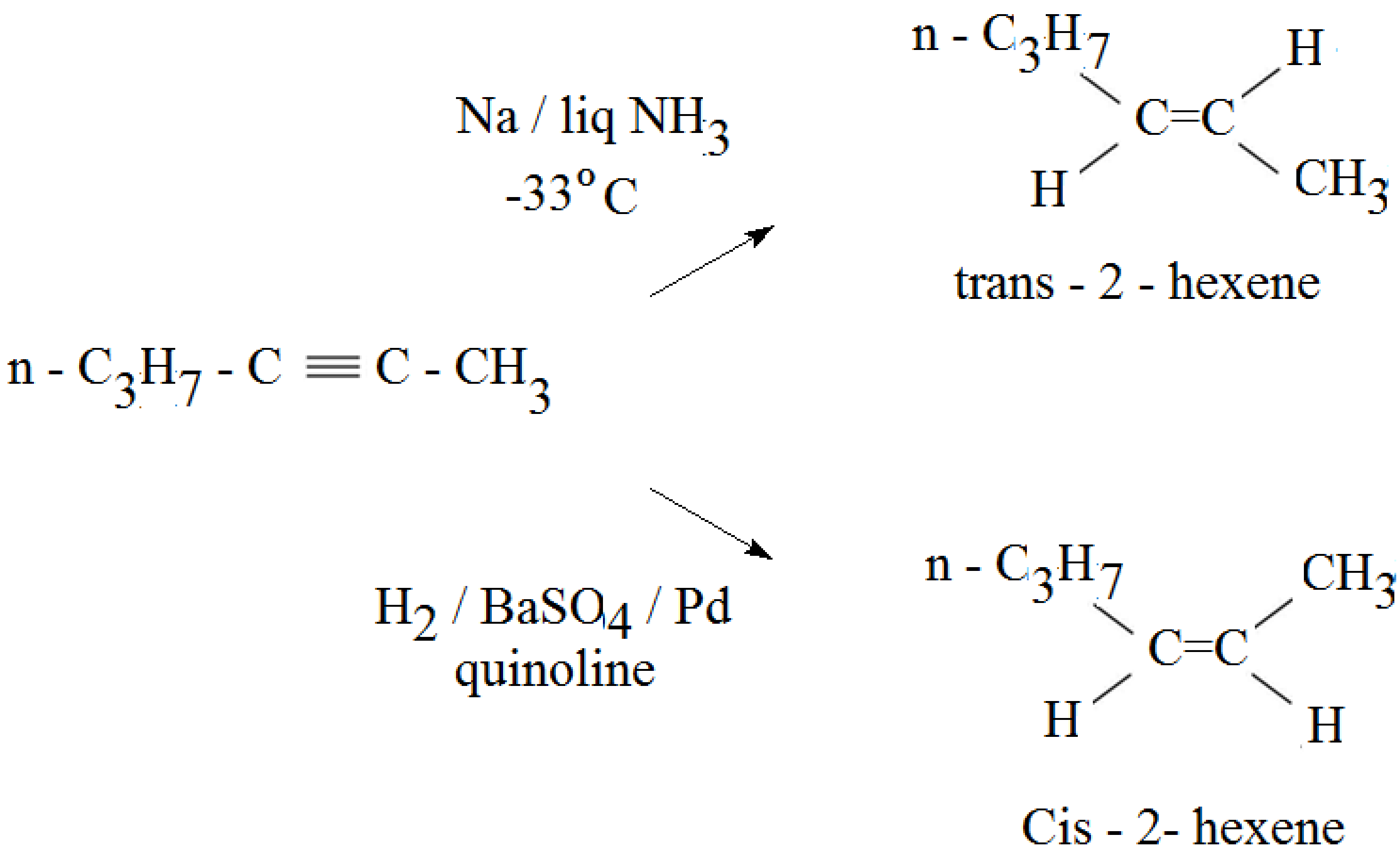
- Add lots of H_2 with metal catalyst (Pd, Pt, or Ni) to reduce alkyne to alkane, completely saturated.



- Use a special catalyst, Lindlar's catalyst to convert an alkyne to a **cis**-alkene (Powdered $BaSO_4$ coated with Pd, poisoned with quinoline).

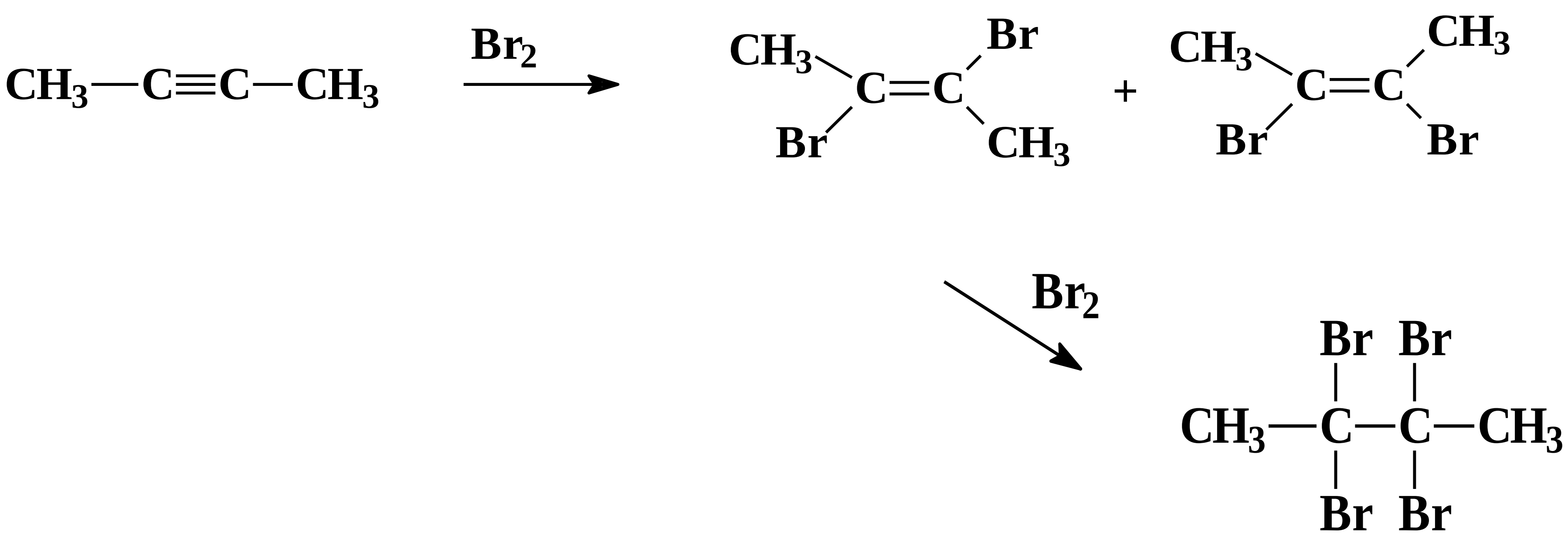


- React the alkyne with sodium in liquid ammonia to form a trans-alkene (Free radical reaction)



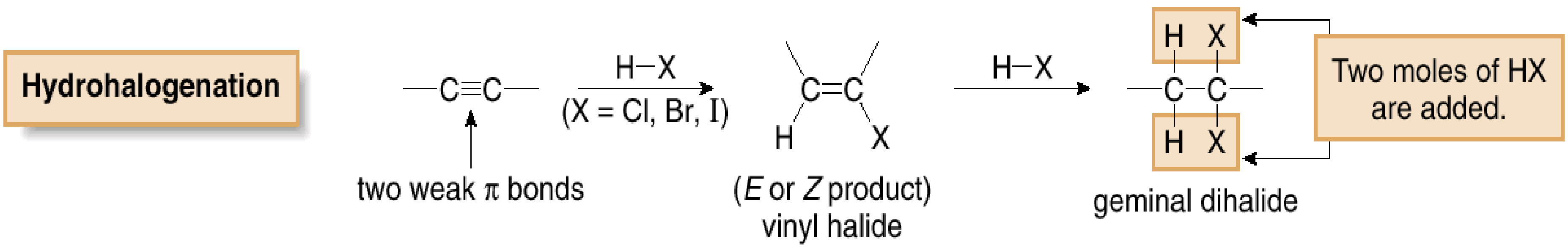
Addition of Halogens

- Cl_2 and Br_2 add to alkynes to form vinyl dihalides.
- May add syn or anti, so product is mixture of **cis** and **trans** isomers.
- Difficult to stop the reaction at dihalide.



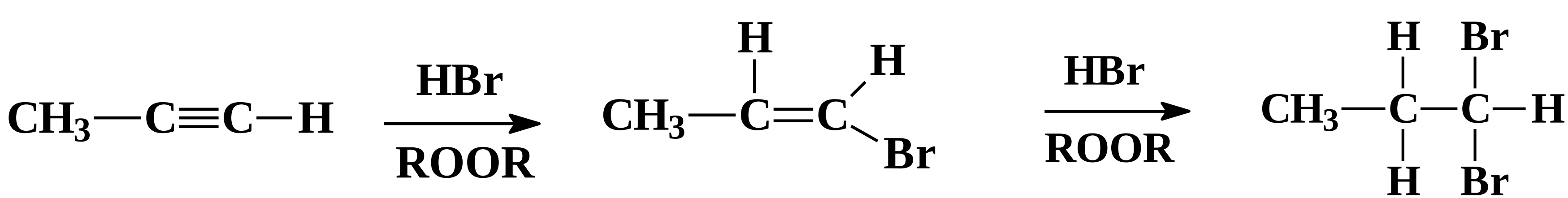
Addition of HX

- HCl, HBr, and HI add to alkynes to form vinyl halides.
- For terminal alkynes, Markovnikov product is formed.
- If two moles of HX is added, product is a geminal dihalide.



HBr with Peroxides

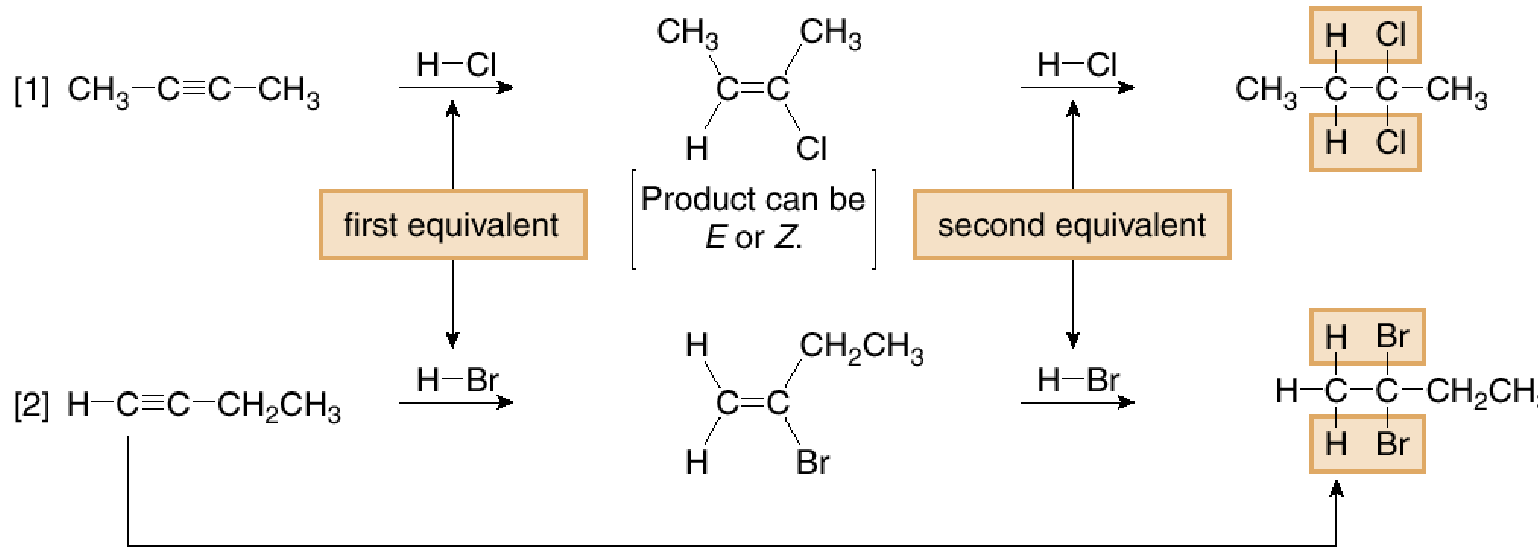
- Anti-Markovnikov product is formed with a terminal alkyne.



Hydrohalogenation: Electrophilic Addition of HX

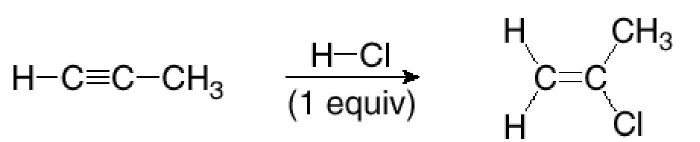
- With two equivalents of HX, both H atoms bond to the *same* carbon.
- With a terminal alkyne, both H atoms bond to the *terminal* carbon; that is, the hydrohalogenation of alkynes follows Markovnikov's rule.

Examples



Both H's end up on the terminal C.

- With only one equivalent of HX, the reaction stops with formation of the vinyl halide.



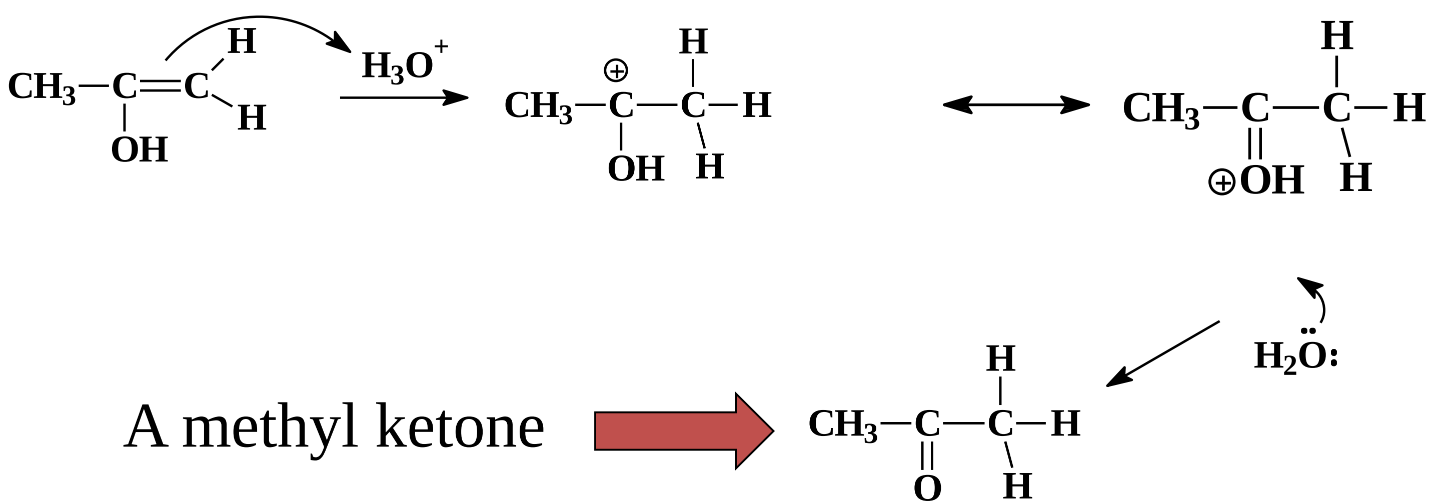
a vinyl chloride
(2-chloropropene)

Hydration of Alkynes

- Mercuric sulfate in aqueous sulfuric acid adds H-OH to one pi bond with a Markovnikov orientation, forming a vinyl alcohol (enol) that rearranges to a ketone.
- Hydroboration-oxidation adds H-OH with an anti-Markovnikov orientation, and rearranges to an aldehyde.

Enol to Keto (in Acid)

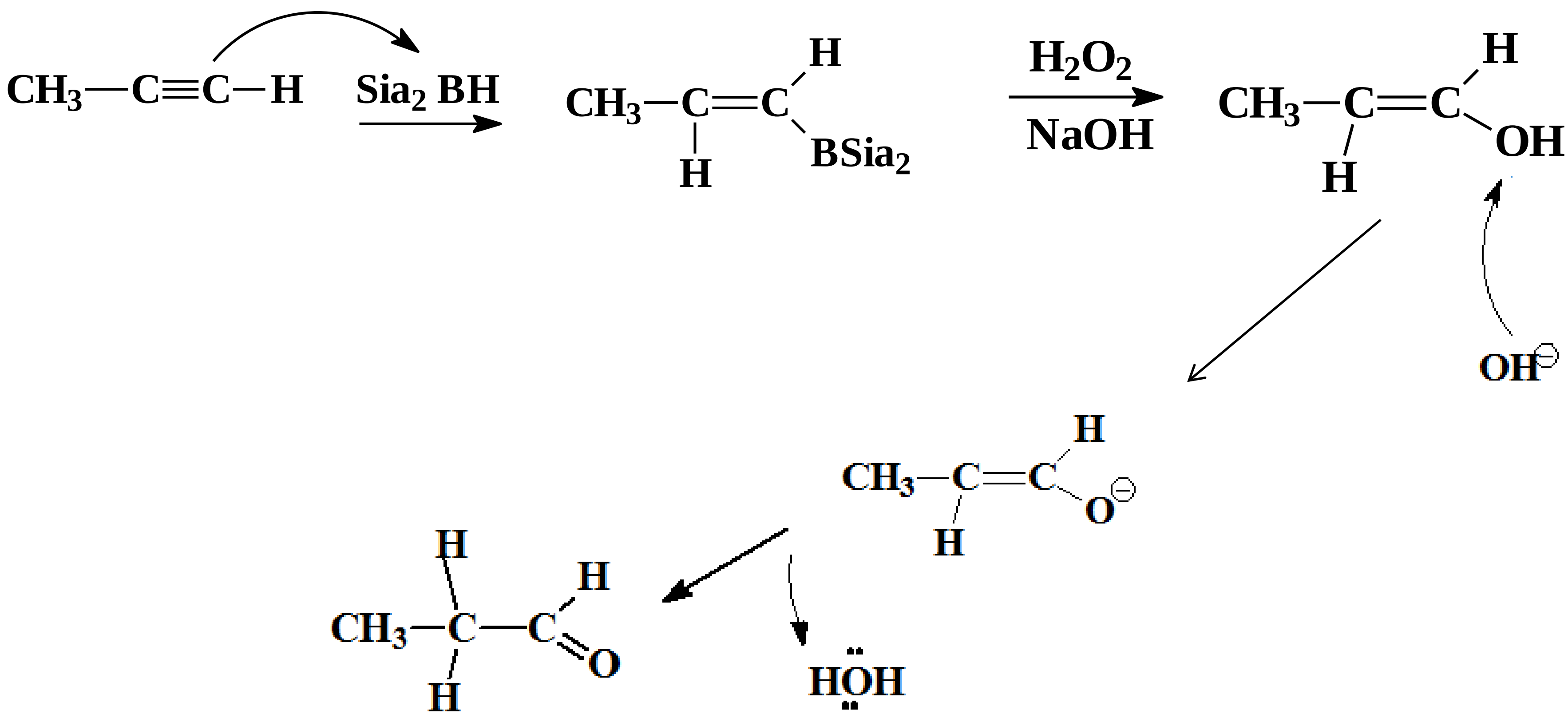
- Add H^+ to the C=C double bond.
- Remove H^+ from OH of the enol.



A methyl ketone

Hydroboration - Oxidation

- B and H of Sia_2BH (Disiamyl borane) add across the triple bond.
- Oxidation with basic H_2O_2 gives the enol.
- H^+ is removed from OH of the enol.
- Then water gives H^+ to the adjacent carbon convert to aldehyde or ketone.

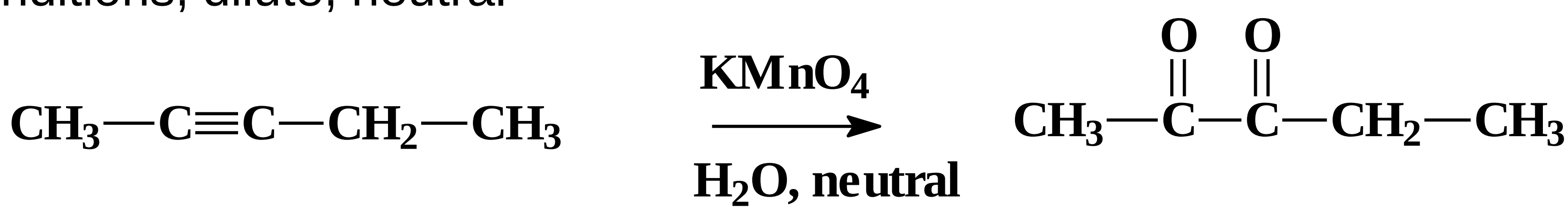


Oxidation of Alkynes

- Similar to oxidation of alkenes.
- Dilute, neutral solution of $KMnO_4$ oxidizes alkynes to a diketone.
- Warm, basic $KMnO_4$ cleaves the triple bond.
- Ozonolysis, followed by hydrolysis, cleaves the triple bond.

Reaction with $KMnO_4$

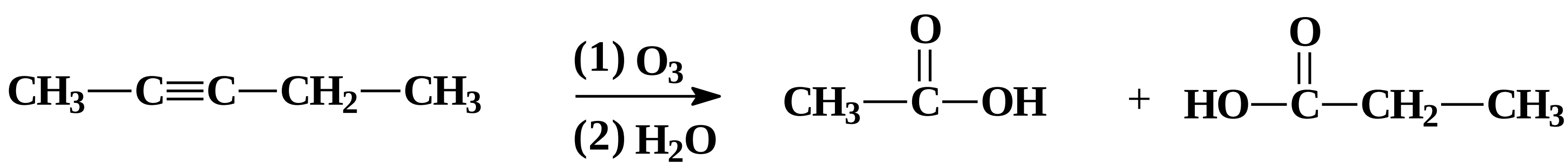
Mild conditions, dilute, neutral



Harsher conditions, warm, basic

Ozonolysis

- Ozonolysis of alkynes produces carboxylic acids (Alkenes gave aldehydes and ketones)



- Used to find location of triple bond in an unknown compound.

Four addition reactions of 1-butyne